

A-level

# ENVIRONMENTAL SCIENCE

## Questions by topic: Marine productivity & fishing - Set 1

### Mark scheme

AQA spec 3.5.2.1-3.5.2.2; Biological resources; textbook pp. 338-351

### Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01.1     | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• In winter nutrients are high (storms disturb sediments, deep mixing) but light is too low for photosynthesis <b>(1)</b></li> <li>• In spring light rises while nutrients are still high so growth is rapid - the spring bloom (peak about May) <b>(1)</b></li> <li>• The bloom uses up nutrients (falling to 10 by July); dead plankton sink out of the photic zone carrying nutrients to the seabed <b>(1)</b></li> <li>• So nutrients become limiting in summer despite high light; a smaller autumn bloom follows when storms return nutrients while light is still adequate <b>(1)</b></li> </ul> | 4     |
| 01.2     | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Warm surface water is stable (no seasonal mixing / storms) so nutrients lost to deep water in sinking dead organisms are not replaced <b>(1)</b></li> <li>• Nutrient (especially phosphate) concentrations in the photic zone remain very low - the limiting factor <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                             | 2     |
| 01.3     | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Upwelling (offshore winds, seamounts, currents deflected upwards) brings cold nutrient-rich deep water into the photic zone <b>(1)</b></li> <li>• Coastal waters receive nutrients washed off the land in rivers and storms disturb shallow sediments <b>(1)</b></li> <li>• Nutrients plus light in the shallow photic zone allow high algal productivity supporting rich food webs and fisheries (eg Peru anchovy) <b>(1)</b></li> </ul>                                                                                                                                                             | 3     |
| 01.4     | <p><i>1 mark.</i></p> <ul style="list-style-type: none"> <li>• Phosphates have low solubility and no atmospheric reservoir; they are only added by river runoff or upwelling and are lost to sediments <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 338-339; AQA spec 3.5.2.1.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 02.1     | <p>2 marks for each point (partial credit for a partly correct point).</p> <ul style="list-style-type: none"> <li>• Demersal trawl: seabed species - cod, haddock, plaice, shrimps, scampi; chains / weights damage the seabed (mix aerobic and anaerobic layers, destroy sea fans, corals, seagrass); high by-catch in mixed fisheries; high fuel use (2)</li> <li>• Purse seine: shoaling pelagic species - tuna, sardines, herring, anchovy; can remove whole shoals; dolphins associated with tuna shoals may be caught (2)</li> <li>• Pelagic long-line: tuna, squid; albatrosses (long-lived, low breeding rate) drown on baited hooks; sharks and turtles caught (2)</li> </ul> | 6     |
| 02.2     | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>• Shrimp trawls use very small mesh so crabs, molluscs, sea urchins, starfish, juvenile fish cannot escape - by-catch can be 99% of the catch (1)</li> <li>• Drift nets are non-selective curtains near the surface catching whales, dolphins, turtles, sharks (1)</li> </ul>                                                                                                                                                                                                                                                                                                                                     | 2     |
| 02.3     | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>• Lost or discarded nets and traps continue to catch and kill organisms; dead animals act as bait attracting more (1)</li> <li>• Biodegradable ropes / panels so lost traps fall apart; radio transmitters on traps so they can be recovered; net recovery schemes (1)</li> </ul>                                                                                                                                                                                                                                                                                                                                 | 2     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 340-343; AQA spec 3.5.2.2.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |
| 03.1     | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>• k-selected: very slow growth, does not breed until 30, few eggs, so the population cannot replace losses (1)</li> <li>• Fish reach catchable size (20) before they breed (30) so fishing can remove the entire future breeding population (1)</li> <li>• Deep-water aggregations easily located and caught by trawling (1)</li> <li>• Recovery takes decades so fisheries have had to be closed (1)</li> </ul>                                                                                                                                                                                                  | 4     |
| 03.2     | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>• Herring breed at 2 years - the same age they become catchable - so there is always a breeding population and many young (1)</li> <li>• Cod are catchable at 2 but do not mature until 4, so heavy fishing removes most fish before they breed (1)</li> </ul>                                                                                                                                                                                                                                                                                                                                                    | 2     |
| 03.3     | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>• Larger numbers of smaller / younger fish are caught, or bigger boats, nets and sonar increase catch effort (1)</li> <li>• Masks the decline of the breeding population until a sudden collapse (1)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                   | 2     |
| 03.4     | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>• Neither is a target species but both are caught in demersal trawls; skate now extinct in the Baltic and critically endangered elsewhere (1)</li> <li>• Very late maturity (skate 11 years, Greenland shark 100-150 years, living over 400) and few young mean even low by-catch mortality exceeds replacement (1)</li> </ul>                                                                                                                                                                                                                                                                                    | 2     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 341-342, 348-349; AQA spec 3.5.2.2.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
| 04.1     | <p>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</p> <ul style="list-style-type: none"> <li>• <math>600 + (90 + 150) - (220 + 80) = 540</math> (1)</li> <li>• Stock fell by 60 so the catch exceeds the MSY / unsustainable (1)</li> </ul>                                                                                                                                                                                                                                                                                                                                            | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>04.2</b> | <i>1 mark for the correct answer.</i><br>• $A + G - M = 90 + 150 - 80 = 160$ <b>(1)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1     |
| <b>04.3</b> | <i>1 mark for each point.</i><br>• Catch is landed and weighed at ports so it is recorded <b>(1)</b><br>• Other terms need data on fish in the sea: populations are mobile (crossing national boundaries), very unevenly distributed <b>(1)</b><br>• Representative sampling of a large enough proportion of the area is impractical / expensive; survival of eggs and larvae varies greatly between years <b>(1)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3     |
| <b>04.4</b> | <i>2 marks for each point (partial credit for a partly correct point).</i><br>• CPUE: if more effort (boat-days, engine power, net size) is needed to land the same catch, the population is declining even though total catch is maintained; a better measure than catch size which technology can prop up <b>(2)</b><br>• Falling mean size / age (from scale or otolith growth rings) shows older adults are being removed faster than replaced and the catch depends on small young fish <b>(2)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 349-351; AQA spec 3.5.2.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |
| <b>05.1</b> | <i>2 marks for each point (partial credit for a partly correct point).</i><br>• Normal diagonal mesh closes as the net fills / stretches so small fish cannot escape; square-mesh panels do not close so juveniles escape to grow and breed; TEDs let turtles out; larger mesh sizes generally <b>(2)</b><br>• Minimum size lets fish reach breeding age before capture (eg UK lobster minimum carapace above breeding size, cod 35 cm); maximum size / V-notching egg-carrying females protects large, highly fecund breeding adults (Hebridean lobsters over 145 mm, Nassau grouper) <b>(2)</b><br>• NTZs (Lundy, Lamlash Bay, Chagos, Great Barrier Reef) protect breeding populations and habitats; surplus young spill over and increase surrounding catches; damaged communities recover <b>(2)</b><br>• Closed seasons, ideally in the breeding season, let fish grow and breed; fishermen may oppose because shoaling breeders are easy to catch <b>(2)</b> | 8     |
| <b>05.2</b> | <i>1 mark for each point.</i><br>• Pelagic species form single-species shoals so fishing stops when the quota is reached <b>(1)</b><br>• In mixed fisheries many species are caught together; once one species' quota is reached it must be discarded (dead) while fishing for others continues <b>(1)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |
| <b>06.1</b> | <i>Any 3 from the points below - 1 mark each, maximum 3.</i><br>• Sinkers / weighted lines so hooks sink quickly below the reach of albatrosses <b>(1)</b><br>• Setting lines at night when birds are not feeding <b>(1)</b><br>• Bird-scaring streamers / decoys <b>(1)</b><br>• Curved hooks that catch tuna but not birds <b>(1)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>06.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Effort: limits on boat size, engine power, days at sea, number of licences; decommissioning payments to reduce fleet size where overfishing is unavoidable <b>(1)</b></li> <li>• Methods: bans or length limits on drift nets; bans on demersal trawling over sensitive seabed (Rockall Bank, Hatton Bank deep-water reefs) <b>(1)</b></li> <li>• Shorter trawl tows so turtles / by-catch survive (over 99% survive under 10 minutes) <b>(1)</b></li> </ul>                                                                                     | 3     |
| <b>06.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Eggs / larvae are reared in captivity through the planktonic stage when most die (predation, failure to find habitat) <b>(1)</b></li> <li>• Juveniles are released when survival chances are good, increasing recruitment to the adult population <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                          | 2     |
| <b>06.4</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Egg / larval survival varies greatly between years so adult numbers do not predict recruitment <b>(1)</b></li> <li>• Several poor years would lower the MSY; egg surveys allow catches to be reduced in advance rather than after a sudden collapse <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                        | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |
| <b>07.1</b> | <p><i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i></p> <ul style="list-style-type: none"> <li>• By-catch = <math>5.9 + 1.9 = 7.8</math> t; total = 8.4 t <b>(1)</b></li> <li>• <math>7.8 / 8.4 \times 100 = 92.9\%</math> (accept 93%) <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                             | 2     |
| <b>07.2</b> | <p><i>1 mark.</i></p> <ul style="list-style-type: none"> <li>• The nets have a very small mesh size so few non-target animals can escape <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1     |
| <b>07.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• They breed at an old age (skate about 11 years; Greenland shark 100-150 years) and produce few young <b>(1)</b></li> <li>• They may reach a catchable size before they are sexually mature, so fishing can remove the whole breeding population before it has bred <b>(1)</b></li> <li>• Populations recover slowly, unlike r-selected species (herring, mackerel) which breed young and produce many young; they are caught as by-catch in demersal trawls so fishing continues even though they have no commercial value <b>(1)</b></li> </ul> | 3     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 07.4     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Impacts: population decline of k-selected species, by-catch (immature target fish, over-quota fish, non-commercial species; drift nets - whales, dolphins, turtles; longlines - albatrosses; pair trawls - porpoises), ghost fishing, seabed damage by demersal trawls, coral and seagrass damage, dynamite fishing, food-web effects</li> <li>• Catch quotas: limit weight landed; work well for single-species pelagic shoals but in mixed fisheries over-quota fish are discarded dead so the quota fails to protect them</li> <li>• Gear design: mesh size and square-mesh escape panels let small fish escape (diagonal mesh closes under drag); turtle exclusion devices; acoustic pingers for dolphins; curved hooks, decoys, sinkers and night setting to protect albatrosses; biodegradable / radio-tracked traps to reduce ghost fishing</li> <li>• Restricted effort: limits on boat size, engine power and days at sea; decommissioning; restricted methods - drift-net bans, demersal trawling bans on sensitive seabeds (Rockall Bank, Hatton Bank)</li> <li>• Evaluation: enforcement at sea is difficult; some methods reduce catch or raise costs; quotas need reliable stock data; effectiveness depends on species (single vs mixed shoals) and on international cooperation as fish migrate; as local stocks decline fishing moves to more distant waters</li> <li>• Justified conclusion weighing effectiveness of different methods</li> </ul> | 9     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 341-345; AQA spec 3.5.2.2.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |